Xingyu Zhao

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Education

University of Wisconsin-Madison Sep 2024 - Present
B.S. in Mathematics and Computer Sciences (Honors)
Expected Dec 2026 | GPA: **4.00/4.00**

Fudan University Sep 2022 - Jun 2024
Undergraduate studies in Computer Science
GPA: **3.90/4.00** | Class rank: **1/88**

Research Interests

Statistical machine learning; information-theoretic analysis of learning dynamics; generalization under distribution shift; statistical structure and geometry of behavioral transfer in language models.

Publications and Manuscripts

[1] **Shattered Compositionality: Counterintuitive Learning Dynamics of Transformers for Arithmetic.**
Xingyu Zhao, Darsh Sharma, Rheeya Uppaal, and Yiqiao Zhong.
NeurIPS 2026, accepted arXiv:2601.22510 | Code

[2] **Cross-Trait Transfer in Subliminal Learning.**
Xingyu Zhao and Yiqiao Zhong.
Under review at ICLR 2027 arXiv:2610.04260 | Code

Research Experience

University of Wisconsin-Madison May 2025 - Present
Undergraduate Researcher, Department of Statistics | Advisor: Prof. Yiqiao Zhong

Learning Dynamics and Generalization in Transformers

- Designed controlled experiments across addition, multiplication, comparison, and sorting, identifying counterintuitive orders of subskill acquisition. In the addition task, models learned output digits from the most to least significant place, opposite to the direction of carry propagation.
- Analyzed the conditional-dependence structure of uniformly sampled addition data to explain this learning order: individual input digits are marginally independent of lower-place output digits, yet same-column input/output pairs become dependent given outgoing carry. Developed conditional mutual-information diagnostics to track subskill acquisition.
- Designed a targeted ablation that randomized only the thousands-place digit of the result; models then failed to learn any lower-place digits, despite unchanged inputs and correct lower-place labels.
- Designed balanced sampling and controlled distribution shifts to isolate subskill interference in comparison and sorting; showed that conflicting comparison cues impair generalization despite strong in-distribution accuracy.
- Trained transformers with up to 100M parameters from scratch; extended robustness evaluation to 10 pretrained LLMs on mathematical word problems with added reasoning steps.

Research Experience *(continued)*

University of Wisconsin-Madison | Advisor: Prof. Yiqiao Zhong | May 2025 - Present

Statistical Structure of Hidden Behavioral Transfer

- Investigated how student language models acquire teacher preferences through semantically unrelated outputs. Formulated directed transfer matrices revealing statistical network structure aligned with hue families and geographic groups of cities.
- Developed six diagnostic metrics based on shifts in model output distributions and internal representation geometry; computed their Pearson correlations with observed transfer matrices.
- Tested whether cross-trait transfer concentrates within semantic groups using permutation tests with Holm correction across 30 experimental settings; found stronger within-group transfer in every setting.
- Applied singular-value analysis to transfer matrices across training checkpoints; increasing effective rank revealed a progression from broad behavioral shifts to more differentiated, trait-specific transfer.
- Fine-tuned 7–8B models across four language-model families using LoRA; separately compared supervised fine-tuning and distribution distillation to assess how training objectives affect transfer.

Selected Coursework

Grades: A in all listed courses.

Advanced mathematics: Analysis I & II; Advanced Linear Algebra; Abstract Algebra; Stochastic Processes; Stochastic Calculus.

The mathematics courses above carry UW-Madison's graduate-coursework designation.

Optimization: Introductory course in linear, integer, and mixed-integer programming.

Theoretical computer science: Algorithms; Data Structures; Theory of Computation.

Machine learning: Deep Learning; Reinforcement Learning.

Systems and graphics: Parallel Programming; Computer Systems; Database Systems; Computer Graphics.

Research Methods and Technical Skills

Statistical methods: Mutual-information diagnostics; conditional-dependence analysis; permutation tests with Holm correction; matrix spectral analysis (SVD, effective rank).

Experimental design: Controlled sampling; ablation studies; evaluation under distribution shift.

Model interpretation: Latent activation analysis; representation geometry; activation steering.

Model training: Transformer training from scratch; supervised fine-tuning with LoRA; distribution distillation.

Scientific computing: Python (PyTorch, Hugging Face Transformers, NumPy, pandas, Matplotlib); C/C++; Git; Linux.

Selected Software Projects

B-Spline Interactive Demo GitHub

Computer Graphics course project | JavaScript

- Implemented an interactive B-spline visualizer using Cox-de Boor recursion, draggable control points, configurable degree and knot vectors, and real-time piecewise-basis visualization.
- Adopted as an interactive learning tool and lecture aid in Prof. Michael Gleicher's CS559 Computer Graphics course.

Fudan University Study Room Booking Prototype GitHub

Django | SQLite | Bootstrap

- Built a full-stack study-room reservation prototype with server-side booking workflows, persistent relational storage, and a responsive user interface.